

Getting started with STM32F4xxxx MCU hardware development

Introduction

This application note is intended for system designers who require an overview of the hardware implementation of the development board, with focus on features like

- power supply
- package selection
- clock management
- reset control
- boot mode settings
- debug management.

This document shows how to use the high-density high-performance microcontrollers listed in *Table 1*, and describes the minimum hardware resources required to develop an application based on those products.

Detailed reference design schematics are also contained in this document, together with descriptions of the main components, interfaces and modes.

Table 1. Applicable products

Type	Part numbers and Product lines
Microcontrollers	STM32F401xB / STM32F401xC
	STM32F401xD / STM32F401xE
	STM32F405/415 line
	STM32F407/417 line
	STM32F410x8 / STM32F410xB
	STM32F411xC / STM32F411xE
	STM32F412xE / STM32F412xG
	STM32F413/423 line
	STM32F427/437 line
	STM32F429/439 line
	STM32F446 line
	STM32F469/479 line

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1 Reference documents

The following documents are available on www.st.com.

Table 2. Referenced documents

Reference	Title
AN2867	Oscillator design guide for ST microcontrollers
AN2606	STM32 microcontroller system memory boot mode
AN3364	Migration and compatibility guidelines for STM32 microcontroller applications

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2 Power supplies

The operating voltage supply (V_{DD}) range is 1.8 V to 3.6 V, which can be reduced down to 1.7 V with some restrictions, as detailed in the product datasheets. An embedded regulator is used to supply the internal 1.2 V digital power.

The real-time clock (RTC), backup registers and backup registers can be powered from the V_{BAT} voltage when the main V_{DD} supply is powered off.

2.1 Digital supply

2.1.1 Voltage regulator

The voltage regulator is always enabled after reset. It works in three different modes depending on the application modes.

- in Run mode, the regulator supplies full power to the 1.2 V domain (core, memories and digital peripherals)
- in Stop mode, the regulator supplies low power to the 1.2 V domain, preserving the contents of the registers and SRAM
- in Standby mode, the regulator is powered down. The contents of the registers and SRAM are lost except for those concerned with the Standby circuitry and the Backup domain.

Note: Depending on the selected package, there are specific pins that should be connected either to V_{SS} or V_{DD} to activate or deactivate the voltage regulator. Refer to section “Voltage regulator” in datasheet for details.

2.1.2 Regulator OFF mode

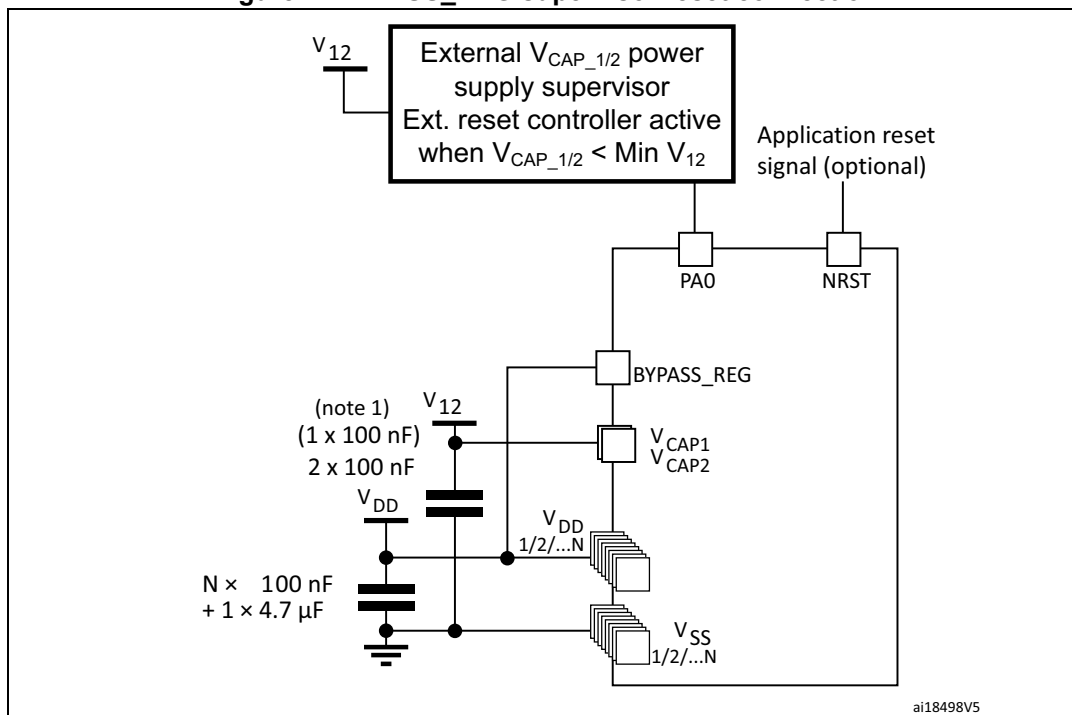
Refer to section “Voltage regulator” in datasheet for details.

- When $BYPASS_REG = V_{DD}$, the core power supply should be provided through V_{CAP1} and V_{CAP2} pins connected together.
 - The two V_{CAP} ceramic capacitors should be replaced by two 100 nF decoupling capacitors.
 - Since the internal voltage scaling is not managed internally, the external voltage value must be aligned with the targeted maximum frequency.
 - When the internal regulator is OFF, there is no more internal monitoring on V12. An external power supply supervisor should be used to monitor the V12 of the logic power domain (V_{CAP}).

PA0 pin should be used for this purpose, and act as power-on reset on V12 power domain.

- In regulator OFF mode, the following features are no more supported:
 - PA0 cannot be used as a GPIO pin since it allows to reset a part of the V12 logic power domain which is not reset by the NRST pin.
 - As long as PA0 is kept low, the debug mode cannot be used under power-on reset. As a consequence, PA0 and NRST pins must be managed separately if the debug connection under reset or pre-reset is required.
 - The over-drive and under-drive modes are not available.
 - The Standby mode is not available.

Figure 1. BYPASS_REG supervisor reset connection



1. V_{CAP2} is not available on all packages. In that case, a single 100 nF decoupling capacitor is connected to V_{CAP1} .

The following conditions must be respected:

- V_{DD} should always be higher than V_{CAP} to avoid current injection between power domains.
- If the time for V_{CAP} to reach V12 minimum value is smaller than the time for V_{DD} to reach 1.7 V, then PA0 should be kept low to cover both conditions: until V_{CAP} reaches V12 minimum value and until V_{DD} reaches 1.7 V.
- Otherwise, if the time for V_{CAP} to reach V12 minimum value is smaller than the time for V_{DD} to reach 1.7 V, then PA0 could be asserted low externally.
- If V_{CAP} goes below V12 minimum value and V_{DD} is higher than 1.7 V, then PA0 must be asserted low externally.

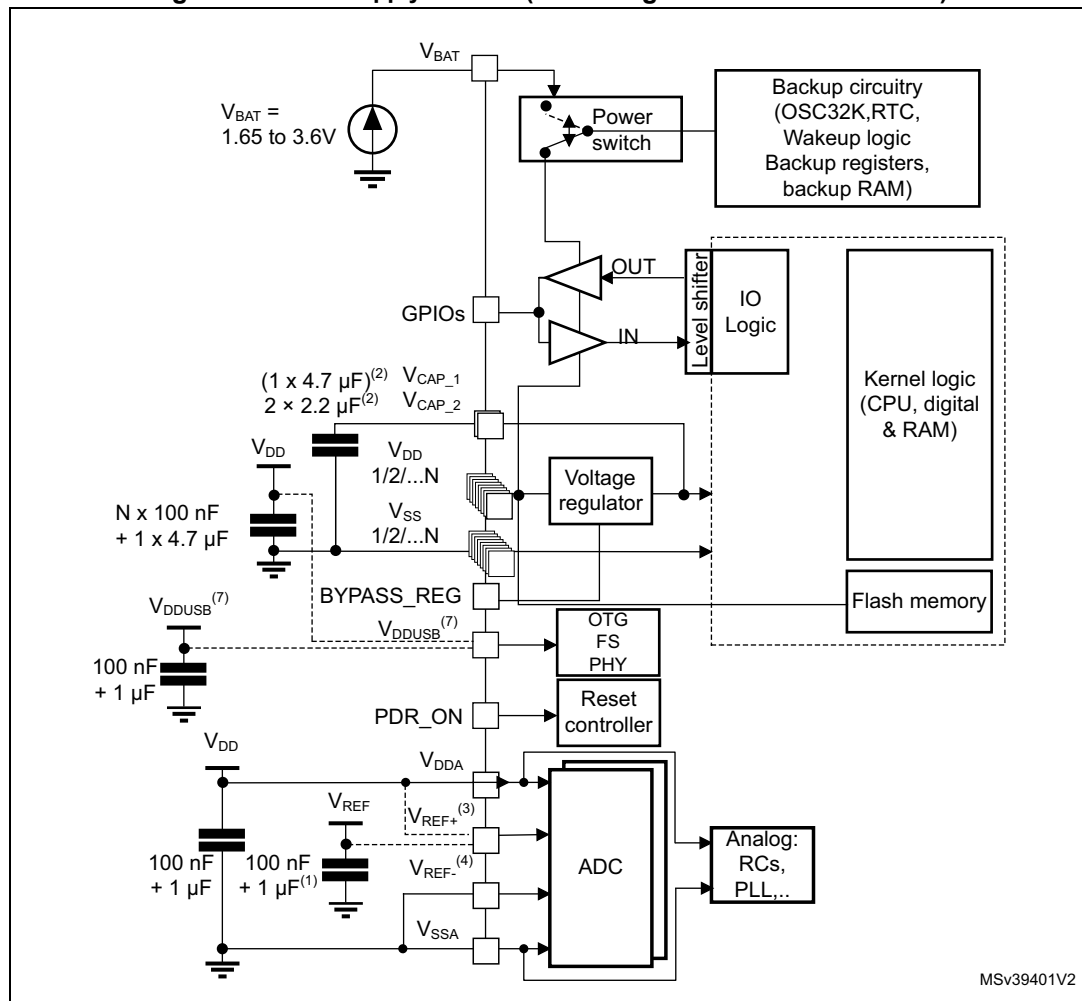
2.2 Power supply schemes

The circuit is powered by a stabilized power supply, V_{DD} .

Caution: The V_{DD} voltage range is 1.8 V to 3.6 V (down to 1.7 V with some restrictions, see relative Datasheet for details).

- The V_{DD} pins must be connected to V_{DD} with external decoupling capacitors: one single Tantalum or Ceramic capacitor (min. 4.7 μ F typ. 10 μ F) for the package + one 100 nF Ceramic capacitor for each V_{DD} pin.
- The V_{BAT} pin can be connected to the external battery (1.65 V < V_{BAT} < 3.6 V). If no external battery is used, it is recommended to connect this pin to V_{DD} with a 100 nF external ceramic decoupling capacitor.
- The V_{DDA} pin must be connected to two external decoupling capacitors (100 nF Ceramic + 1 μ F Tantalum or Ceramic).
- The V_{REF+} pin can be connected to the V_{DDA} external power supply. If a separate, external reference voltage is applied on V_{REF+} , a 100 nF and a 1 μ F capacitors must be connected on this pin. In all cases, V_{REF+} must be kept between (V_{DDA} -1.2 V) and V_{DDA} with minimum of 1.7 V.
- Additional precautions can be taken to filter analog noise:
 - V_{DDA} can be connected to V_{DD} through a ferrite bead.
 - The V_{REF+} pin can be connected to V_{DDA} through a resistor.
- For the voltage regulator configuration, there is specific BYPASS_REG pin (not available on all packages) that should be connected either to V_{SS} or V_{DD} to activate or deactivate the voltage regulator specific.
 - Refer to [Section 2.1.2](#) and section "Voltage regulator" of the related device datasheet for details.
- When the voltage regulator is enabled, V_{CAP1} and V_{CAP2} pins must be connected to 2*2.2 μ F LowESR < 2 Ω Ceramic capacitor (or 1*4.7 μ F LowESR < 1 Ω Ceramic capacitor if only V_{CAP1} pin is provided on some packages).

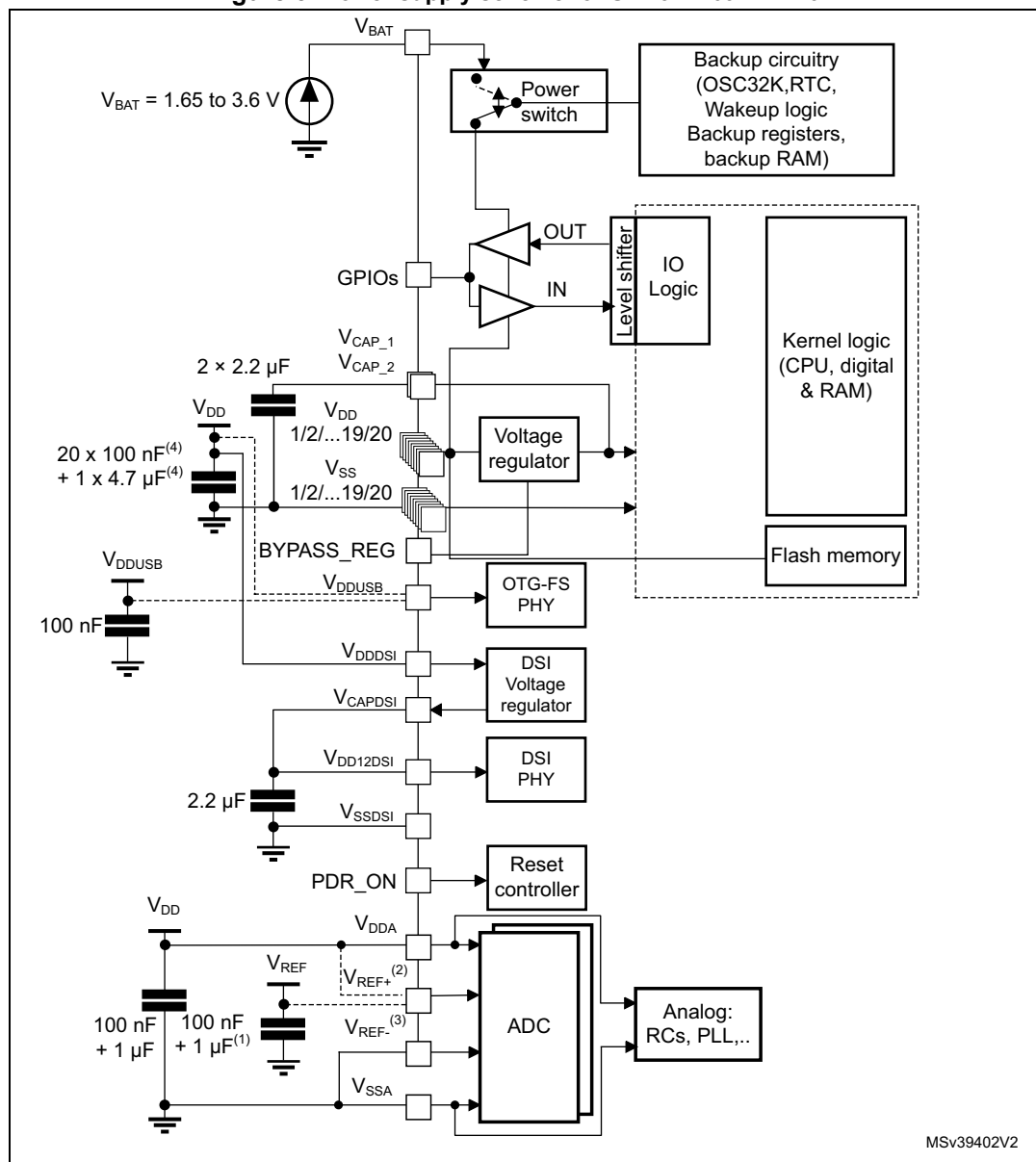
Figure 2. Power supply scheme (excluding STM32F469xx/F479xx)



MSv39401V2

1. Optional. If a separate, external reference voltage is connected on V_{REF+} , the two capacitors (100 nF and 1 μ F) must be connected.
2. V_{CAP2} is not available on all packages. In that case, a single 4.7 μ F (ESR < 1 Ω) is connected to V_{CAP1} .
3. V_{REF+} is either connected to V_{REF+} or to V_{DDA} (depending on package).
4. V_{REF-} is either connected to V_{REF-} or to V_{SSA} (depending on package).
5. N is the number of V_{DD} and V_{SS} inputs.
6. Refer to datasheet for BYPASS_REG and PDR_ON pins connection.
7. V_{DDUSB} is only available on STM32F446xx.
8. Backup RAM is not available on STM32F410xx, STM32F411xx, STM32F412xx, STM32F413xx, and STM32F423xx.

Figure 3. Power supply scheme for STM32F469xx/F479xx



MSv39402V2

1. Optional. If a separate, external reference voltage is connected on V_{REF+}, the two capacitors (100 nF and 1 μF) must be connected.
2. V_{REF+} is either connected to V_{REF+} or to V_{DDA} (depending on package).
3. V_{REF-} is either connected to V_{REF-} or to V_{SSA} (depending on package).
4. Refer to datasheet for BYPASS_REG and PDR_ON pins connection.

2.3 Analog Supply

To improve conversion accuracy, the ADC has an independent power supply that can be filtered separately, and shielded from noise on the PCB.

- The ADC voltage supply input is available on V_{DDA} pin.
- An isolated supply ground connection is provided on the V_{SSA} pin.

In all cases, the V_{SSA} pin should be externally connected to same supply ground than V_{SS} .

To ensure a better accuracy on low-voltage inputs, the user can connect a separate external reference voltage ADC input on V_{REF+} . The voltage on V_{REF+} may range from ($V_{DDA} - 1.2$ V) to V_{DDA} with a minimum of 1.7 V.

When available (depending on package), V_{REF-} must be externally tied to V_{SSA} .

3 Reset and power supply supervisor

3.1 System reset

A system reset sets all registers to their reset values except for the reset flags in the clock controller CSR register and the registers in the Backup domain (see [Figure 2](#)).

A system reset is generated when one of the following events occurs:

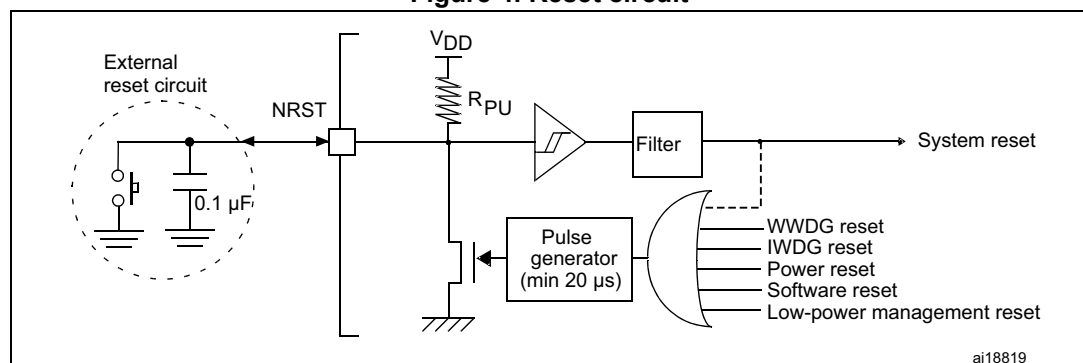
1. A low level on the NRST pin (external reset)
2. window watchdog end-of-count condition (WWDG reset)
3. Independent watchdog end-of-count condition (IWDG reset)
4. A software reset (SW reset)
5. Low-power management reset

The reset source can be identified by checking the reset flags in the Control/Status register, RCC_CSR.

The products listed in [Table 1](#) do not require an external reset circuit to power-up correctly. Only a pull-down capacitor is recommended to improve EMS performance by protecting the device against parasitic resets, as exemplified in [Figure 4](#).

Charging and discharging a pull-down capacitor through an internal resistor increases the device power consumption. The capacitor recommended value (100 nF) can be reduced to 10 nF to limit this power consumption.

Figure 4. Reset circuit



3.1.1 NRST circuitry example

This example applies to STM32F410xx, STM32F411xx, STM32F412xx, STM32F413xx, STM32F423xx, STM32F446xx and STM32F469 where PDR_ON can be connected to V_{SS} to permanently disable internal reset circuitry.

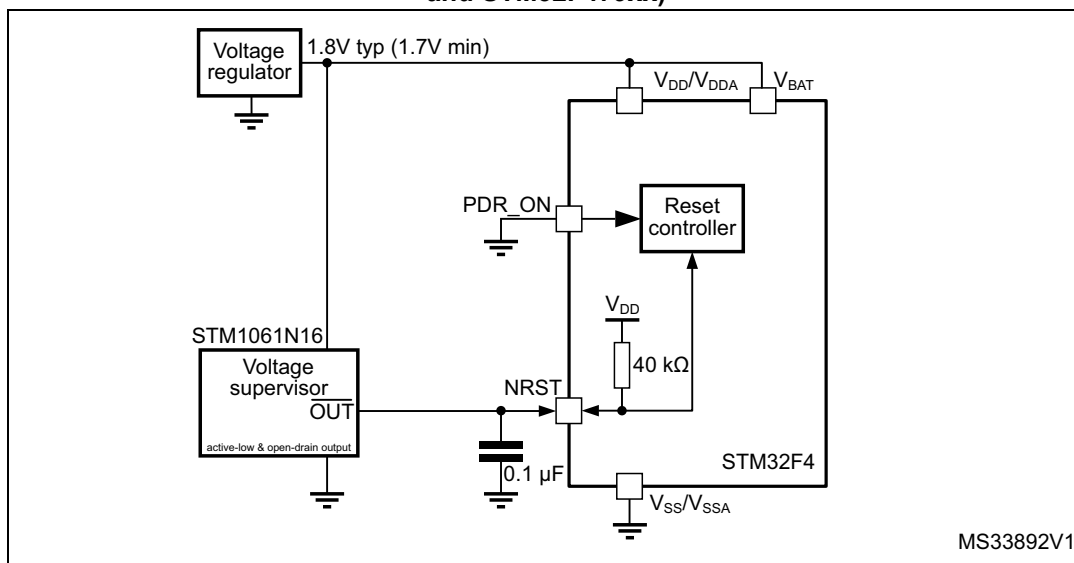
Restrictions:

- PDR_ON = 0 is mostly intended for V_{DD} supply between 1.7 V and 1.9V (i.e. 1.8V +/- 5% supply). Supply ranges which never go below 1.8V minimum should be better managed by

internal circuitry (no additional component needed, thanks to fully embedded reset controller).

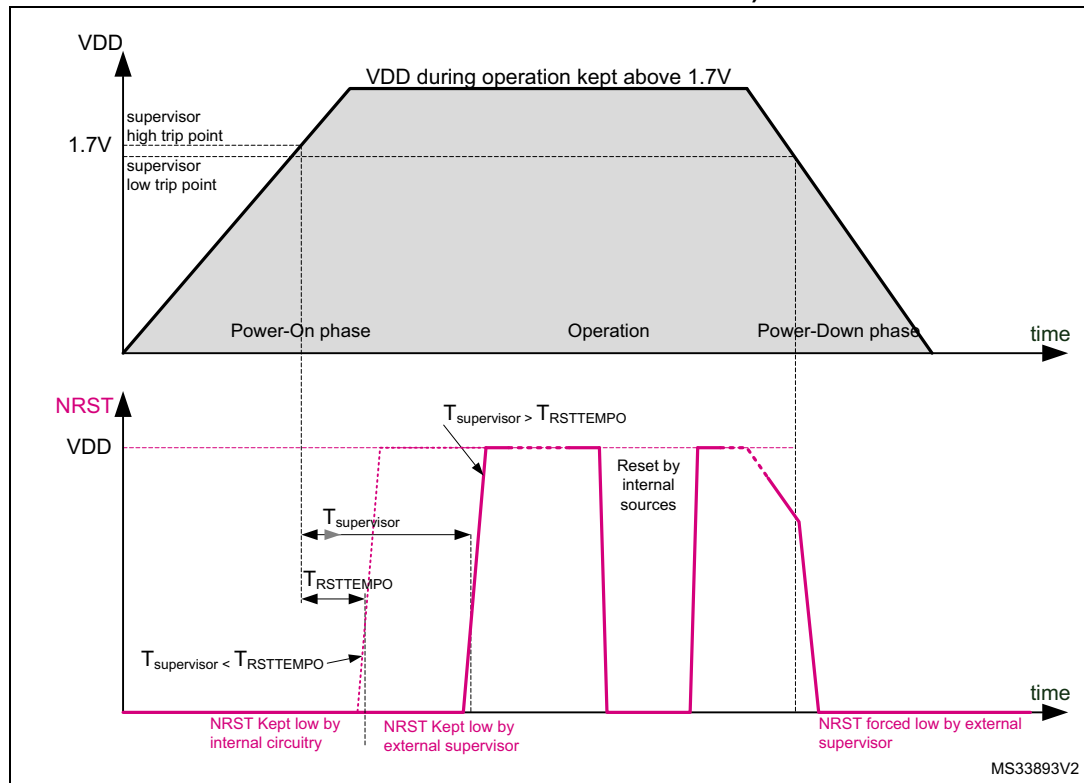
- When the internal reset is OFF, the following integrated features are no longer supported:
 - The integrated power-on reset (POR) / power-down reset (PDR) circuitry is disabled.
 - The brownout reset (BOR) circuitry must be disabled.
 - The embedded programmable voltage detector (PVD) is disabled.
 - VBAT functionality is no more available and V_{BAT} pin should be connected to V_{DD} .

Figure 5. NRST circuitry example (only for STM32F410xx, STM32F411xx, STM32F412xx, STM32F413xx, STM32F423xx, STM32F446xx, STM32F469xx and STM32F479xx)



Even with $PDR_ON=0$, during power up, the NRST is driven low by internal Reset controller during $T_{RSTTEMPO}$ in order to allow stabilization of internal analog circuitry. Refer to STM32F4xxxx datasheets for actual timing value.

Figure 6. NRST circuitry timings example (not to scale, only for STM32F410xx, STM32F411xx, STM32F412xx, STM32F413xx, STM32F423xx, STM32F446xx, STM32F469xx and STM32F479xx)



Selection of NRST voltage supervisor

Voltage supervisor should have the following characteristics

- Reset output **active-low open-drain** (output driving low when voltage is below trip point).
 - Supervisor trip point including tolerances and hysteresis should fit the expected V_{DD} range.
Notice that supervisor spec usually specify trip point for falling supply, so hysteresis should be added to check the power on phase.

3.2 Power supply supervisor

3.2.1 PDR_ON circuitry example

Note: This example doesn't apply to STM32F410xx, STM32F411xx, STM32F412xx, STM32F413xx, STM32F423xx, STM32F446xx, STM32F469xx and STM32F479xx, where PDR_ON can be connected to VSS to permanently disable internal reset circuitry (external voltage supervisor required on NRST pin). Thanks to backward compatibility, circuitry built for other STM32F4xxxx products will work for STM32F410xx, STM32F411xx, STM32F412xx, STM32F413xx, STM32F423xx, STM32F446xx, STM32F469xx and STM32F479xx.

Note: Please contact your local STMicroelectronics representative or visit www.st.com in case you want to use circuitry different from the one described hereafter.

Restrictions:

- PDR_ON = 0 is mostly intended for V_{DD} supply between 1.7 V and 1.9V (i.e. 1.8V +/- 5% supply).
Supply ranges which never go below 1.8V minimum should be better managed with internal circuitry (no additional component thanks to fully embedded reset controller).
- To ensure safe power down, the external voltage supervisor (or equivalent) is required to drive PDR_ON=1 during power off sequence.

When the internal reset is OFF, the following integrated features are no longer supported:

- The integrated power-on reset (POR) / power-down reset (PDR) circuitry is disabled.
- The brownout reset (BOR) circuitry must be disabled.
- The embedded programmable voltage detector (PVD) is disabled.
- V_{BAT} functionality is no more available and V_{BAT} pin should be connected to V_{DD} .

Figure 7. PDR_ON simple circuitry example (not needed for STM32F410xx, STM32F411xx, STM32F413xx, STM32F423xx, STM32F412xx, STM32F446xx, STM32F469xx and STM32F479xx)

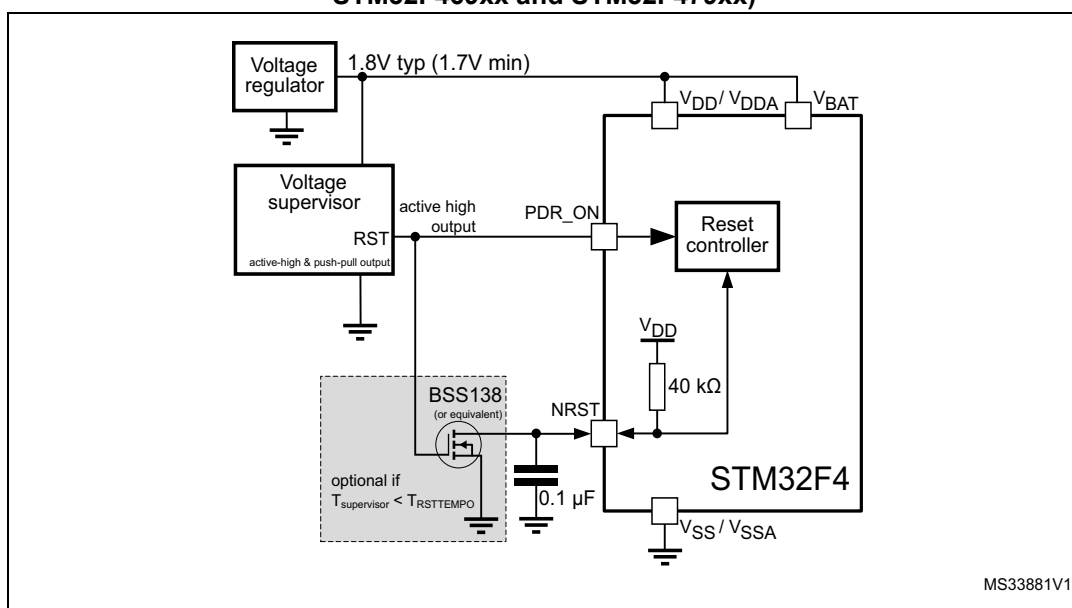
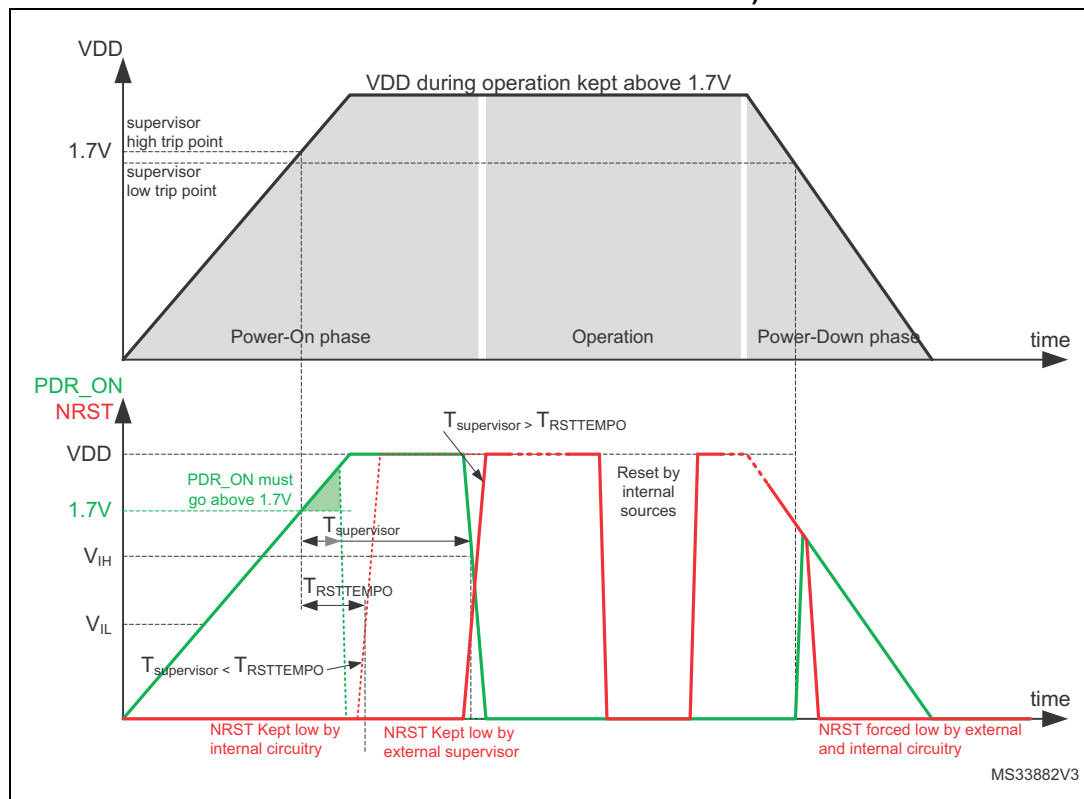


Figure 8. PDR_ON timings example (not to scale, (not needed for STM32F410xx, STM32F411xx, STM32F412xx, STM32F413xx, STM32F423xx, STM32F446xx, STM32F469xx and STM32F479xx))



Selection of PDR_ON voltage supervisor

Voltage supervisor should have the following characteristics

- Reset output **active-high push-pull** (output driving high when voltage is below trip point)
- Supervisor trip point including tolerances and hysteresis should fit the expected V_{DD} range.

Notice that supervisor spec usually specify trip point for falling supply, so hysteresis should be added to check the power on phase.

Example:

- Voltage regulator 1.8V +/- 5% mean V_{DD} min 1.71V
- Supervisor specified at 1.66V +/- 2.5% with an hysteresis of 0.5% mean
 - rising trip max = 1.71V (1.66V + 2.5% + 0.5%)
 - falling trip min = 1.62V (1.66V - 2.5%).

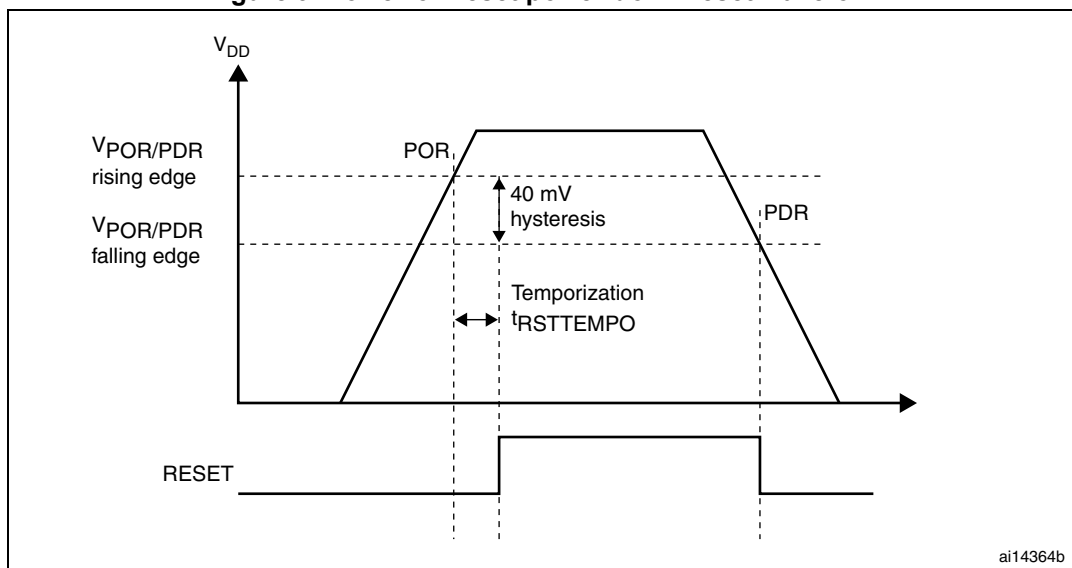
3.2.2 Power on reset (POR) / power down reset (PDR)

The device has an integrated POR/PDR circuitry that allows proper operation starting from 1.8 V.

The device remains in the Reset mode as long as V_{DD} is below a specified threshold, $V_{POR/PDR}$, without the need for an external reset circuit. For more details concerning the

power on/power down reset threshold, refer to the electrical characteristics in the product datasheets.

Figure 9. Power-on reset/power-down reset waveform



1. $t_{RSTTEMPO}$ is approximately 2.6 ms. $V_{POR/PDR}$ rising edge is 1.74 V (typ.) and $V_{POR/PDR}$ falling edge is 1.70 V (typ.). Refer to STM32F4xxx datasheets for actual value.

The internal power-on reset (POR) / power-down reset (PDR) circuitry is disabled through the PDR_ON pin. An external power supply supervisor should monitor V_{DD} and should maintain the device in reset mode as long as V_{DD} is below a specified threshold. PDR_ON should be connected to this external power supply supervisor. See [Section 3.2.1](#) for details.

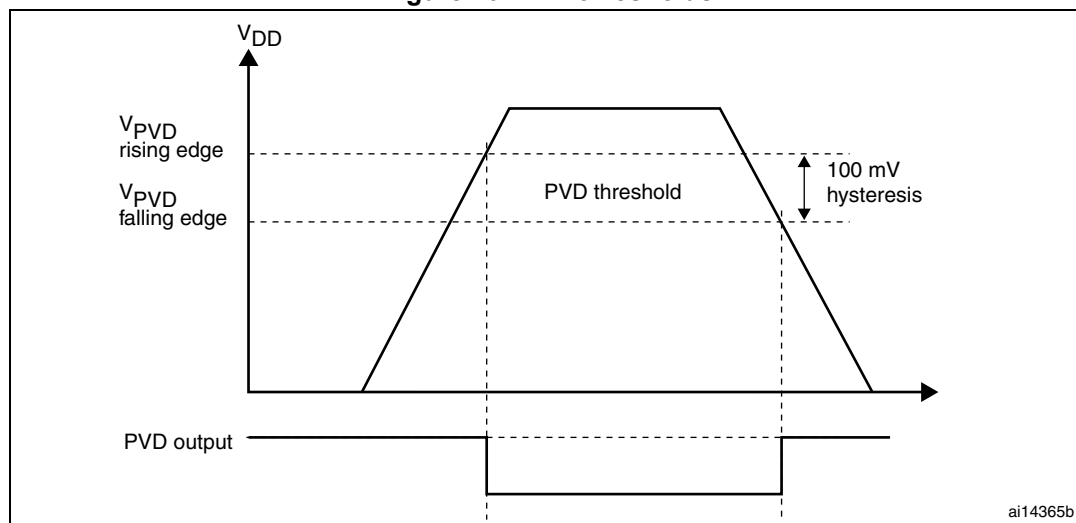
3.2.3 Programmable voltage detector (PVD)

You can use the PVD to monitor the V_{DD} power supply by comparing it to a threshold selected by the PLS[2:0] bits in the Power control register (PWR_CR).

The PVD is enabled by setting the PVDE bit.

A PVDO flag is available, in the Power control/status register (PWR_CSR), to indicate whether V_{DD} is higher or lower than the PVD threshold. This event is internally connected to EXTI Line16 and can generate an interrupt if enabled through the EXTI registers. The PVD output interrupt can be generated when V_{DD} drops below the PVD threshold and/or when V_{DD} rises above the PVD threshold depending on the EXTI Line16 rising/falling edge configuration. As an example the service routine can perform emergency shutdown tasks.

Figure 10. PVD thresholds



4 Package

4.1 Package Selection

Package should be selected by taking into account the constraints that are strongly dependent upon the application.

The list below summarizes the more frequent ones:

- Amount of interfaces required.
Some interfaces might not be available on some packages.
Some interfaces combinations could not be possible on some packages.
- PCB technology constraints.
Small pitch and high ball density could require more PCB layers and higher class PCB
- Package height
- PCB available area
- Noise emission or signal integrity of high speed interfaces.
Smaller packages usually provide better signal integrity. This is further enhanced as Small pitch and high ball density requires multilayer PCBs which allow better supply/ground distribution.
- Compatibility with other devices.

Table 3 summarizes the available packages for all STM32F4xxxx family.

Table 3. Package summary

Size (mm) ⁽¹⁾	10 x 10	14 x 14	20 x 20	24 x 24	28 x 28	13 x 13	7 x 7	7 x 7	7 x 7	10 x 10	7 x 7	10 x 10	2.553 x 2.579	3.034 x 3.220	3.060 x 3.060	3.658 x 3.686	4.039 x 3.951	5.693 x 3.815	4.223 x 4.223	4.521 x 5.547	4.891 x 5.692
Sales numbers	LQFP64	LQFP100	LQFP144	LQFP176	LQFP208	TFBGA216	UFQFPN48	UFBGA100	UFBGA144	UFBGA144	UFBGA169	UFBGA176	WLCSP36	WLCSP49	WLCSP49	WLCSP64	WLCSP81	WLCSP81	WLCSP90	WLCSP143	WLCSP168
STM32F401xB/C STM32F401xD/E	X	X					X	X							X						
STM32F405xx/407xx	X	X	X	X								X							X		
STM32F410xx	X						X						X								
STM32F411xx	X	X					X	X						X							
STM32F412xx	X	X	X				X	X		X						X					
STM32F413xx/423xx	X	X	X				X	X		X							X				
STM32F415xx/417xx	X	X	X	X								X							X		
STM32F427xx/429xx		X	X	X	X	X					X	X								X	

Table 3. Package summary (continued)

Size (mm) ⁽¹⁾	10 x 10	14 x 14	20 x 20	24 x 24	28 x 28	13 x 13	7 x 7	7 x 7	7 x 7	10 x 10	7 x 7	10 x 10	2.553 x 2.579	3.034 x 3.220	3.060 x 3.060	3.658 x 3.686	4.039 x 3.951	5.693 x 3.815	4.223 x 4.223	4.521 x 5.547	4.891 x 5.692
Sales numbers	LQFP64	LQFP100	LQFP144	LQFP176	LQFP208	TFBGA216	UFQFPN48	UFBGA100	UFBGA144	UFBGA144	UFBGA169	UFBGA176	WLCSP36	WLCSP49	WLCSP49	WLCSP64	WLCSP81	WLCSP81	WLCSP90	WLCSP143	WLCSP168
STM32F437xx/439xx		X	X	X	X	X					X	X								X	
STM32F446XX	X	X	X						X	X								X			
STM32F469xx				X	X	X					X	X									X
STM32F479xx				X	X	X					X	X									X

1. body size, excluding pins

4.2 Pinout Compatibility

4.2.1 I/O speed

- When using the GPIO as I/O, design considerations have to be taken into account to ensure that the operation is as intended.
- When the load capacitance becomes larger, the rise/fall time of the I/O pin increases. This capacitance includes the effects of the board traces.
- I/O characteristics are available in the product's datasheet.
- [Table 4](#) illustrates the I/O AC characteristics of the STM32F469xx:

Table 4. I/O AC characteristics⁽¹⁾⁽²⁾

OSPEEDR y[1:0] bit value ⁽¹⁾	Symbol	Parameter	Conditions	Min	Typ	Max	Unit
00	$f_{\max(\text{IO})\text{out}}$	Maximum frequency	$C_L = 50 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	4	MHz
			$C_L = 50 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	2	
			$C_L = 10 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	8	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.8 \text{ V}$	-	-	4	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	3	
	$t_{f(\text{IO})\text{out}}/$ $t_{r(\text{IO})\text{out}}$	Output high to low level fall time and output low to high level rise time	$C_L = 50 \text{ pF}, V_{DD} = 1.7 \text{ V}$ to 3.6 V	-	-	100	ns
01	$f_{\max(\text{IO})\text{out}}$	Maximum frequency ⁽³⁾	$C_L = 50 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	25	MHz
			$C_L = 50 \text{ pF}, V_{DD} \geq 1.8 \text{ V}$	-	-	12.5	
			$C_L = 50 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	10	
			$C_L = 10 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	50	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.8 \text{ V}$	-	-	20	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	12.5	
	$t_{f(\text{IO})\text{out}}/$ $t_{r(\text{IO})\text{out}}$	Output high to low level fall time and output low to high level rise time	$C_L = 50 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	10	ns
			$C_L = 10 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	6	
			$C_L = 50 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	20	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	10	

Table 4. I/O AC characteristics⁽¹⁾⁽²⁾ (continued)

OSPEEDRy[1:0] bit value ⁽¹⁾	Symbol	Parameter	Conditions	Min	Typ	Max	Unit
10	$f_{\max(\text{IO})\text{out}}$	Maximum frequency ⁽³⁾	$C_L = 40 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	50 ⁽³⁾	MHz
			$C_L = 10 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	100 ⁽³⁾	
			$C_L = 40 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	25	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.8 \text{ V}$	-	-	50	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	42.5	
	$t_{f(\text{IO})\text{out}}/$ $t_{r(\text{IO})\text{out}}$	Output high to low level fall time and output low to high level rise time	$C_L = 40 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	6	ns
			$C_L = 10 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	4	
			$C_L = 40 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	10	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	6	
11	$f_{\max(\text{IO})\text{out}}$	Maximum frequency ⁽³⁾	$C_L = 30 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	100 ⁽³⁾	MHz
			$C_L = 30 \text{ pF}, V_{DD} \geq 1.8 \text{ V}$	-	-	50	
			$C_L = 30 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	42.5	
			$C_L = 10 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	180 ⁽³⁾	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.8 \text{ V}$	-	-	100	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	72.5	
	$t_{f(\text{IO})\text{out}}/$ $t_{r(\text{IO})\text{out}}$	Output high to low level fall time and output low to high level rise time	$C_L = 30 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	4	ns
			$C_L = 30 \text{ pF}, V_{DD} \geq 1.8 \text{ V}$	-	-	6	
			$C_L = 30 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	7	
			$C_L = 10 \text{ pF}, V_{DD} \geq 2.7 \text{ V}$	-	-	2.5	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.8 \text{ V}$	-	-	3.5	
			$C_L = 10 \text{ pF}, V_{DD} \geq 1.7 \text{ V}$	-	-	4	
-	$t_{\text{EXTI}pw}$	Pulse width of external signals detected by the EXTI controller	-	10	-	-	ns

1. Guaranteed by design.

2. The I/O speed is configured using the OSPEEDRy[1:0] bits. Refer to the STM32F4xx reference manual for a description of the GPIOx_SPEEDR GPIO port output speed register.

3. To minimize impact of process variations, IO compensation cell can be activated to reduce the overshoot during rise/fall transitions, for maximum frequencies above 50 MHz and $V_{DD} > 2.4 \text{ V}$.



4.3 Alternate Function

Each pin of the MCU can be configured to multiple functions. These functions are selected by software.

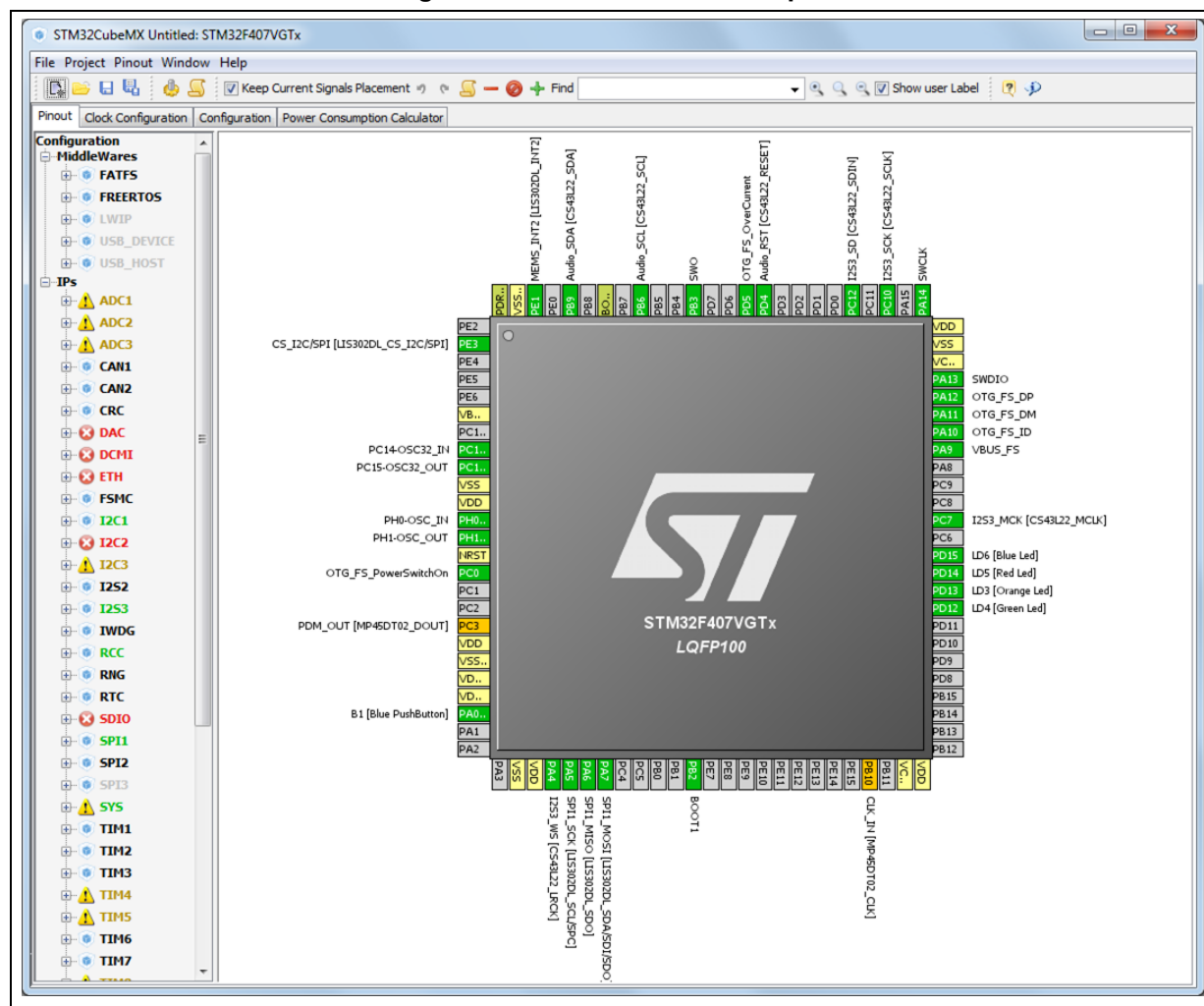
The full description of I/O alternate functions is described in the datasheet of the selected device and the register, which allow the pin configuration, are described in the reference manual.

Table 5. Alternate function

Port		AF0	AF1	AF2	AF3	AF4	AF5	AF6	AF7	AF8	AF9	AF10	AF11	AF12	AF13	AF14	AF15
		SYS	TIM1/2	TIM3/4/5	TIM8/9/10/11	I2C1/2/3	SPI1/2/3/4/5/6	SPI2/3/SAI1	SPI2/3/USART1/2/3	USART6/UART4/5	CAN1/2/TIM12/13/14/QUAD	QUADSPI/OTG2_HS/OTG1_F	ETH	FMC/SDIO/OTG2_	DCMI/DSI HOST	LCD	SYS
Port B	PB0	-	TIM1_CH2N	TIM3_CH3	TIM8_CH2N	-	-	-	-	-	LCD_R3	OTG_HS_ULPI_D1	ETH_MII_RXD2	-	-	LCD_G1	EVENT OUT
	PB11	-	TIM2_CH4	-	-	I2C2_SDA	-	-	USART3_RX	-	-	OTG_HS_ULPI_D4	ETH_MII_TX_EN/ETH_RMII_TX_EN	-	DSI_HOST_TE	LCD_G5	EVENT OUT
	PB12	-	TIM1_BKIN	-	-	I2C2_SMBus	SPI2_NSS/I2S2_WS	-	USART3_CK	-	CAN2_RX	OTG_HS_ULPI_D5	ETH_MII_TXD0/ETH_RMII_TXD0	OTG_HS_ID	-	-	EVENT OUT
	PB13	-	TIM1_CH1N	-	-	-	SPI2_SCK/I2S2_CK	-	USART3_CTS	-	CAN2_TX	OTG_HS_ULPI_D6	ETH_MII_TXD1/ETH_RMII_TXD1	-	-	-	EVENT OUT
	PB14	-	TIM1_CH2N	-	TIM8_CH2N	-	SPI2_MISO	I2S2ext_SD	USART3_RTS	-	TIM12_CH1	-	-	OTG_HS_DM	-	-	EVENT OUT
	PB15	RTC_REFIN	TIM1_CH3N	-	TIM8_CH3N	-	SPI2_MOSI/I2S2_SD	-	-	-	TIM12_CH2	-	-	OTG_HS_DP	-	-	EVENT OUT

In order to easily explore Peripheral Alternate Functions mapping to pins, it is recommended to use the STM32CubeMX tool available on www.st.com.

Figure 11. STM32CubeMX example screen-shot



4.3.1 Handling unused pins

All microcontrollers are designed for a variety of applications and often a particular application does not use 100% of the MCU resources.

To increase EMC performance, unused clocks, counters or I/Os, should not be left free, e.g. I/Os should be set to “0” or “1” (pull-up or pull-down to the unused I/O pins.) and unused features should be “frozen” or disabled.

Note: *To reduce leakage it is advisable to configure the I/O as an analog input or to push-pull and to set it to “0”.*

4.4 Boot mode selection

4.4.1 Boot mode selection

In the STM32F4xxx, three different boot modes can be selected by means of the BOOT[1:0] pins as shown in [Table 6](#).

Table 6. Boot modes

BOOT mode selection pins		Boot mode	Aliasing
BOOT1	BOOT0		
x	0	Main Flash memory	Main Flash memory is selected as boot space
0	1	System memory	System memory is selected as boot space
1	1	Embedded SRAM	Embedded SRAM is selected as boot space

The values on the BOOT pins are latched on the 4th rising edge of SYSCLK after a reset. It is up to the user to set the BOOT1 and BOOT0 pins after reset to select the required boot mode.

Boot from User Flash mode

The application code that runs after reset is located in user flash memory.

The user flash memory in this mode is aliased to start at address 0x00000000 in boot memory space. Upon reset, the top-of-stack value is fetched from address 0x00000000, and code then begins execution at address 0x00000004.

Boot from System Memory mode

The system memory (not the user flash) is now aliased to start at address 0x00000000. The application code in this case must have already been loaded into system memory.

Boot from Embedded SRAM mode

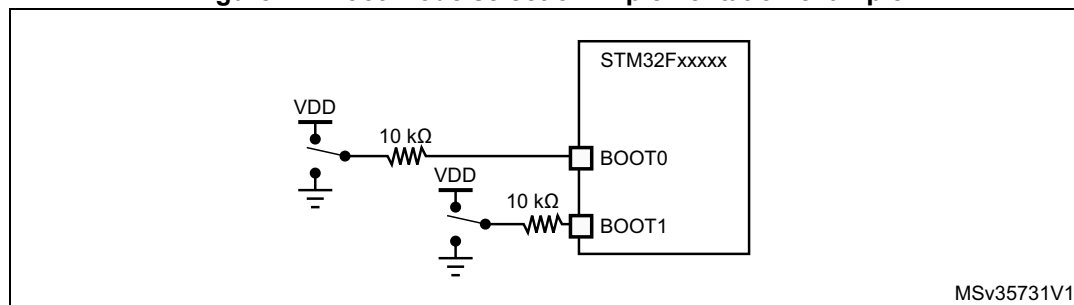
The SRAM start at address 0x00000000. When this mode is selected, the device expects the vector table to have been relocated using the NVIC exception table and offset register, and execution begins at the start of embedded SRAM. The application code in this case must have already been loaded into embedded SRAM.

This last mode is usually used for Debugging.

4.5 Boot pin connection

Figure 12 shows the external connection required to select the boot memory of the STM32F4xxxx.

Figure 12. Boot mode selection implementation example



1. Resistor values are given only as a typical example.

4.6 Embedded boot loader mode

The embedded boot loader is located in the System memory and is programmed by ST during production.

It is used to reprogram the Flash memory using one of the following serial interfaces.

The following table shows the supported communication peripherals by the system bootloader.

Table 7. STM32F4xxxx bootloader communication peripherals

Bootloader peripherals	STM32F401xB/C STM32F401xD/E	STM32F405/415 STM32F407/417 STM32F427/437 STM32F429/439	STM32F410xx	STM32F411xC/ STM32F411xE	STM32F412xx/ STM32F413xx/ STM32F423xx	STM32F469xx/ STM32F479xx
DFU	USB OTG FS (PA11/12) in Device mode	USB OTG FS (PA11/12) in Device mode	-	USB OTG FS (PA11/12) in Device mode	USB OTG FS (PA11/12) in Device mode	USB OTG FS (PA11/12) in Device mode
USART1	PA9/PA10	PA9/PA10	PA9/PA10	PA9/PA10	PA9/PA10	PA9/PA10
USART2	PD5/PD6	-	-	PD5/PD6	PD5/PD6	-
USART3	-	PB10/PB11/ PC10/PC11	-	-	PB10/PB11	PB10/PB11, PC10/PC11
CAN	-	PB5/PB13	-	-	PB5/PB13	PB5/PB13
I2C1	PB6/PB7	-	PB6/PB7	PB6/PB7	PB6/PB7	-
I2C2	PB3/PB10	-	PB3/PB10	PB3/PB10	PF0/PF1	-
I2C3	PA8/PB4	-	-	PA8/PB4	PA8/PB4	-
I2C FMP1	-	-	-	-	PB14/PB15	-
SPI1	PA4/PA5/ PA6/PA7	-	PA4/PA5/ PA6/PA7	PA4/PA5/ PA6/PA7	PA4/PA5/ PA6/PA7	-
SPI2	PB12/PB13/ PB14/PB15	-	PB12/PB13/ PB14/PB15	PB12/PB13/ PB14/PB15	-	-

Table 7. STM32F4xxxx bootloader communication peripherals (continued)

Bootloader peripherals	STM32F401xB/C STM32F401xD/E	STM32F405/415 STM32F407/417 STM32F427/437 STM32F429/439	STM32F410xx	STM32F411xC/ STM32F411xE	STM32F412xx/ STM32F413xx/ STM32F423xx	STM32F469xx/ STM32F479xx
SPI3	PA15/PC10/ PC11/PC12	-	-	PA15/PC10/ PC11/PC12	PA15/PC10/ PC11/PC12	-
SPI4	-	-	-	-	PE11/PE12/ PE13/PE14	-

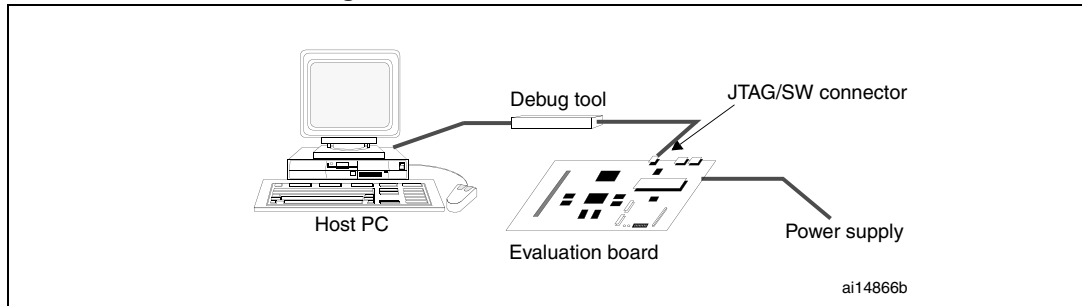
For additional information, refer to AN2606 ([Table 2](#)).

5 Debug management

The Host/Target interface is the hardware equipment that connects the host to the application board. This interface is made of three components: a hardware debug tool, a JTAG or SW connector and a cable connecting the host to the debug tool.

[Figure 13](#) shows the connection of the host to the evaluation board.

Figure 13. Host-to-board connection



5.1 SWJ debug port (serial wire and JTAG)

The STM32F4xxxx core integrates the serial wire / JTAG debug port (SWJ-DP). It is an Arm® standard CoreSight™ debug port that combines a JTAG-DP (5-pin) interface and a SW-DP (2-pin) interface.

- The JTAG debug port (JTAG-DP) provides a 5-pin standard JTAG interface to the AHP-AP port
- The serial wire debug port (SW-DP) provides a 2-pin (clock + data) interface to the AHP-AP port

In the SWJ-DP, the two JTAG pins of the SW-DP are multiplexed with some of the five JTAG pins of the JTAG-DP.

For more details on the SWJ debug port refer to the reference manual of the product, SWJ debug port section (serial wire and JTAG).

5.2 Pinout and debug port pins

The STM32F4xxxx MCU is offered in various packages with different numbers of available pins. As a result, some functionality related to the pin availability may differ from one package to another.

5.2.1 SWJ debug port pins

Five pins are used as outputs for the SWJ-DP as *alternate functions* of general-purpose I/Os (GPIOs). These pins, shown in [Table 8](#), are available on all packages.

Table 8. Debug port pin assignment

SWJ-DP pin name	JTAG debug port		SW debug port		Pin assignment
	Type	Description	Type	Debug assignment	
JTMS/SWDIO	I	JTAG test mode selection	I/O	Serial wire data input/output	PA13
JTCK/SWCLK	I	JTAG test clock	I	Serial wire clock	PA14
JTDI	I	JTAG test data input	-	-	PA15
JTDO/TRACESWO	O	JTAG test data output	-	TRACESWO if async trace is enabled	PB3
JNTRST	I	JTAG test nReset	-	-	PB4

5.2.2 Internal pull-up and pull-down resistors on JTAG pins

The JTAG input pins must *not* be floating since they are directly connected to flip-flops to control the debug mode features. Special care must be taken with the SWCLK/TCK pin that is directly connected to the clock of some of these flip-flops.

To avoid any uncontrolled I/O levels, the STM32F4xxxx embeds internal pull-up and pull-down resistors on JTAG input pins:

- JNTRST: Internal pull-up
- JTDI: Internal pull-up
- JTMS/SWDIO: Internal pull-up
- TCK/SWCLK: Internal pull-down

Once a JTAG I/O is released by the user software, the GPIO controller takes control again. The reset states of the GPIO control registers put the I/Os in the equivalent state:

- JNTRST: Input pull-up
- JTDI: Input pull-up
- JTMS/SWDIO: Input pull-up
- JTCK/SWCLK: Input pull-down
- JTDO: Input floating

The software can then use these I/Os as standard GPIOs.

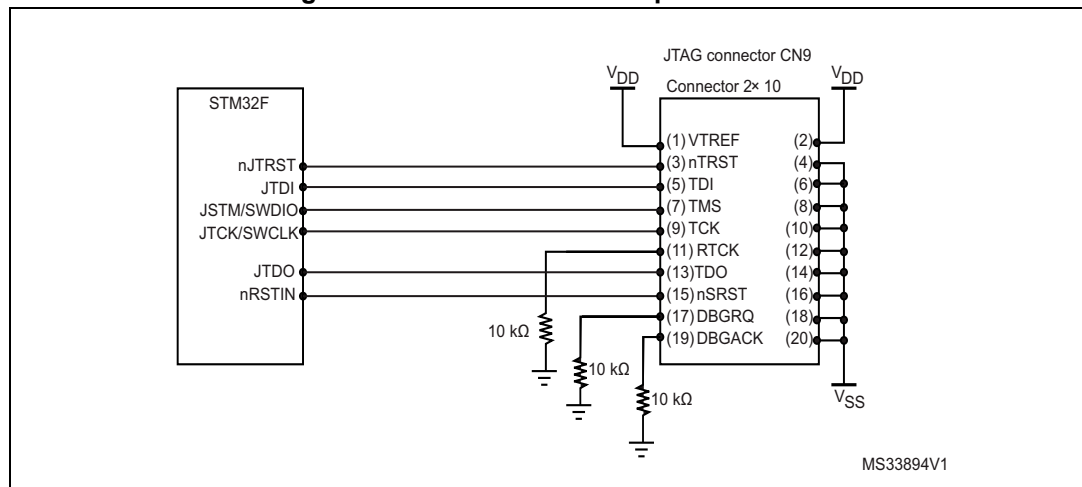
Note: The JTAG IEEE standard recommends to add pull-up resistors on TDI, TMS and nTRST but there is no special recommendation for TCK. However, for the STM32F4xxxx, an integrated pull-down resistor is used for JTCK.

Having embedded pull-up and pull-down resistors removes the need to add external resistors.

5.2.3 SWJ debug port connection with standard JTAG connector

Figure 14 shows the connection between the STM32F4xxxx and a standard JTAG connector.

Figure 14. JTAG connector implementation



6 Clocks

Three different clock sources can be used to drive the system clock (SYSCLK):

- HSI oscillator clock (high-speed internal clock signal)
- HSE oscillator clock (high-speed external clock signal)
- PLL clock

The devices have two secondary clock sources:

- 32 kHz low-speed internal RC (LSI RC) that drives the independent watchdog and, optionally, the RTC used for Auto-wakeup from the Stop/Standby modes.
- 32.768 kHz low-speed external crystal (LSE crystal) that optionally drives the real-time clock (RTCCLK)

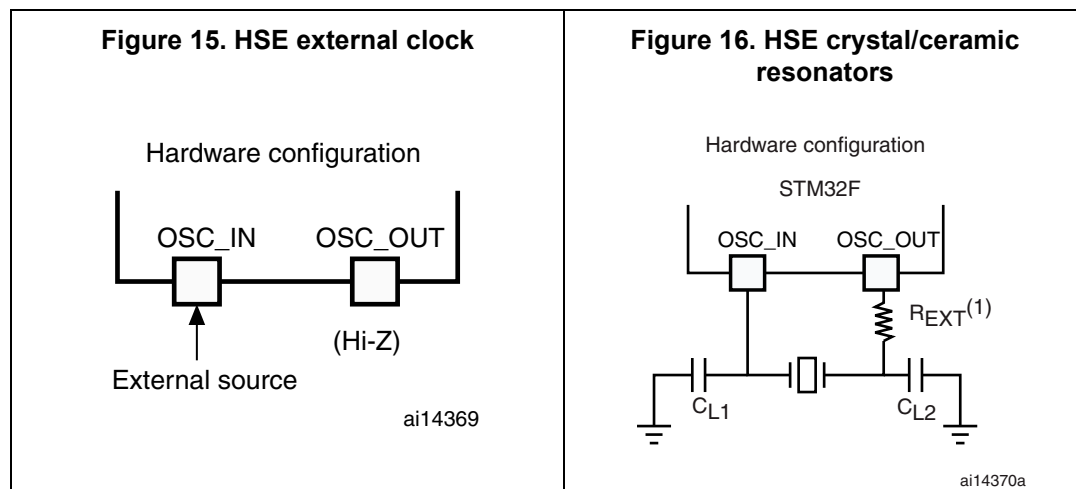
Each clock source can be switched on or off independently when it is not used, to optimize the power consumption.

Refer to the reference manual for the description of the clock tree.

6.1 HSE OSC clock

The high-speed external clock signal (HSE) can be generated from two possible clock sources:

- HSE user external clock (see [Figure 15](#))
- HSE external crystal/ceramic resonator (see [Figure 16](#))



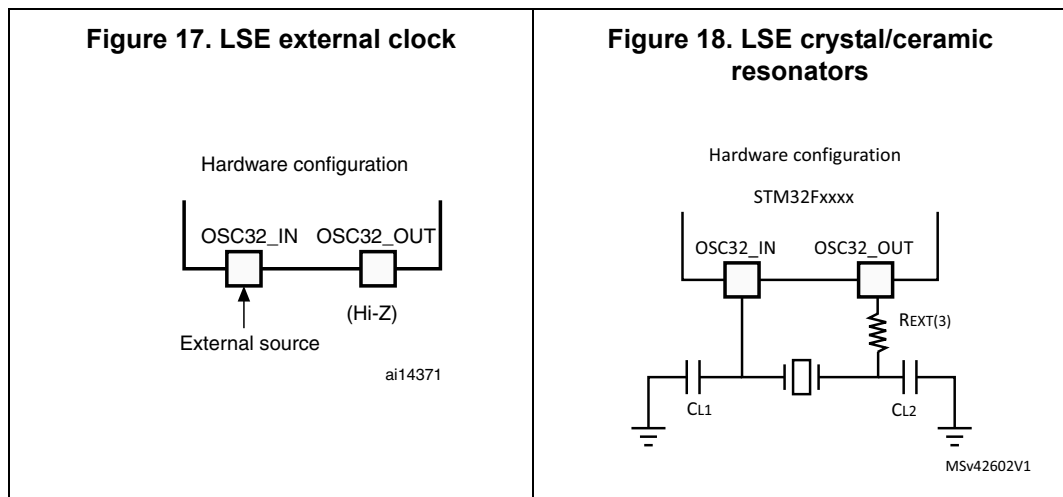
1. The value of R_{EXT} depends on the crystal characteristics. Typical value is in the range of 5 to 6 R_S (resonator series resistance).

Refer to the dedicated Application Note (AN2867 - Oscillator design guide for ST microcontrollers) and electrical characteristics sections in the datasheet of your product for more details.

6.2 LSE OSC clock

The low-speed external clock signal (LSE) can be generated from two possible clock sources:

- LSE user external clock (see [Figure 17](#))
- LSE external crystal/ceramic resonator (see [Figure 18](#))



1. **“LSE crystal/ceramic resonators” figure:**
To avoid exceeding the maximum value of C_{L1} and C_{L2} (15 pF) it is strongly recommended to use a resonator with a load capacitance $C_L \leq 7$ pF. Never use a resonator with a load capacitance of 12.5 pF.
2. **“LSE external clock” and “LSE crystal/ceramic resonators” figures:**
OSC32_IN and OSC32_OUT pins can be used also as GPIO, but it is recommended not to use them as both RTC and GPIO pins in the same application.
3. **“LSE crystal/ceramic resonators” figure:**
The value of R_{EXT} depends on the crystal characteristics. A 0 Ω resistor would work but would not be optimal. To fine tune R_S value, refer to AN2867 - Oscillator design guide for ST microcontrollers ([Table 2](#)) and electrical characteristics sections in the datasheet of your product for more details.

Figure 19. Reference schematic

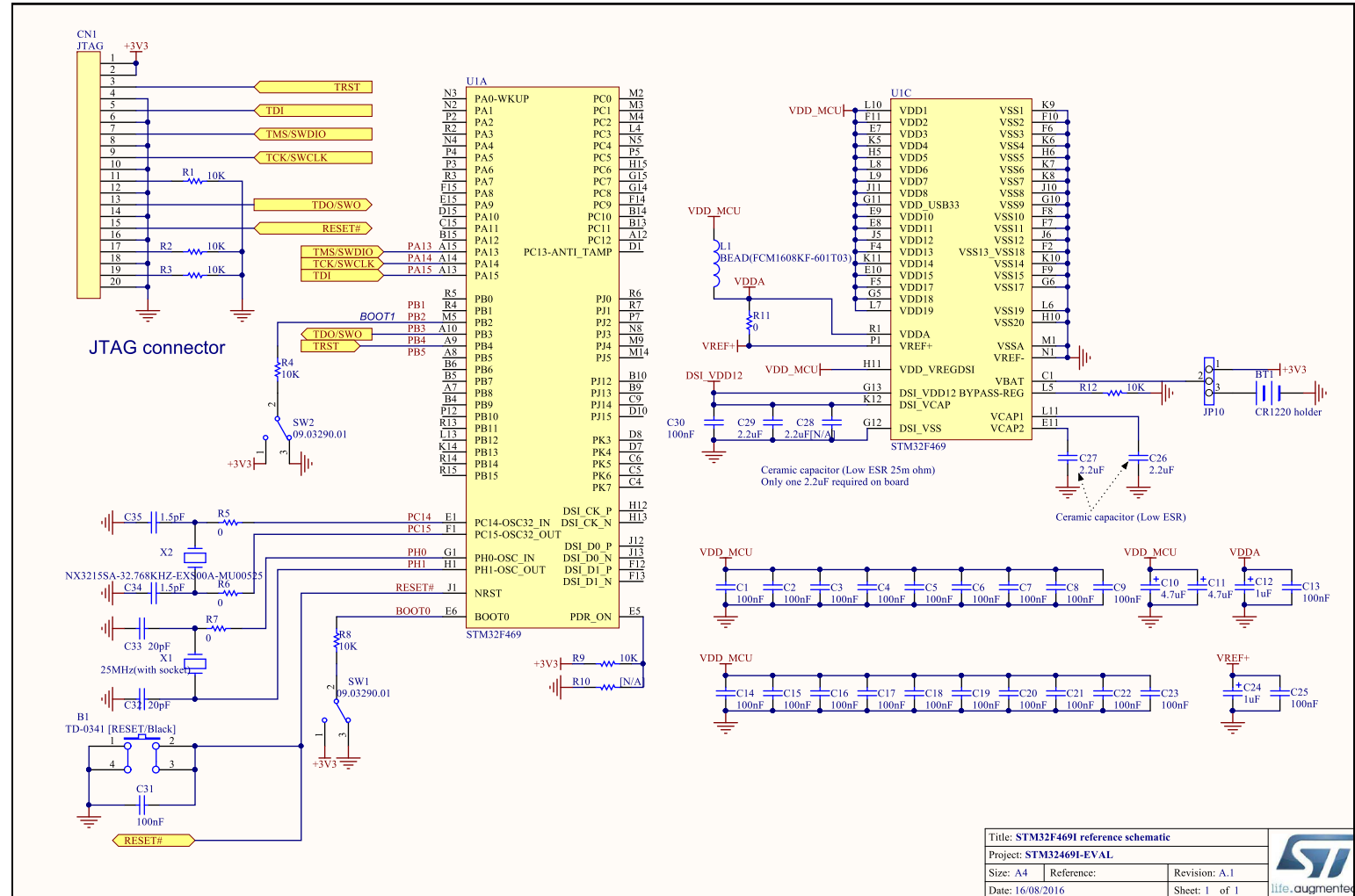


Figure 20. Bill of Material

Comment	Description	Designator	Footprint	Quantity
TD-0341 [RESET/Black]	SE PUSHBUTTON	B1	PB10	1
CR1220 holder	Battery	BT1	BAT_2SM_CR1220	1
100nF	Capacitor	C1, C2, C3, C4, C5, C6, C7, C8, C9, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C25, C30	0402C	22
4.7uF	Polarized Capacitor (Radial)	C10, C11	TAN-A	2
1uF	Polarized Capacitor (Radial)	C12, C24	TAN-A	2
2.2uF	Capacitor	C26, C27	1206C	2
2.2uF[N/A]	Capacitor	C28	1206C	1
2.2uF	Capacitor	C29	0402C	1
100nF	Capacitor	C31	0603C	1
20pF	Capacitor	C32, C33	0603C	2
1.5pF	Capacitor	C34, C35	0603C	2
JTAG		CN1	IDC20S	1
		JP10	SIP3	1
BEAD(FCM1608KF-601T03)	Inductor	L1	0603L	1
10K	Resistor	R1, R2, R3, R4, R8, R9, R12	0603R	7
0	Resistor	R5, R6, R7, R11	0603R	4
[N/A]	Resistor	R10	0603R	1
09.03290.01	SPDT Subminiature Toggle Switch, Right Angle Mounting, Vertical Actuation	SW1, SW2	SW1_3TH_2R54_10X2R5	2
STM32F469		U1	BGA216_0R8_13X13_SKT	1
25MHz(with socket)	Crystal	X1	XTAL_socket	1
NX3215SA-32.768KHZ-EXS00A	Crystal	X2	XTAL_2SM_3R2X1R5	1

8 Recommended PCB routing guidelines for STM32F4xxxx devices

8.1 PCB stack-up

In order to reduce the reflections on high speed signals, it is necessary to match the impedance between the source, sink and transmission lines. The impedance of a signal trace depends on its geometry and its position with respect to any reference planes.

The trace width and spacing between differential pairs for a specific impedance requirement is dependent on the chosen PCB stack-up. As there are limitations in the minimum trace width and spacing which depend on the type of PCB technology and cost requirements, a PCB stack-up needs to be chosen which allows all the required impedances to be realized.

The minimum configuration that can be used is 4 or 6 layers stack-up. An 8 layers boards may be required for a very dense PCBs that have multiple SDRAM/SRAM/NOR/LCD components.

The following stack-ups are intended as examples which can be used as a starting point for helping in a stack-up evaluation and selection. These stack-up configurations are using a GND plane adjacent to the power plane to increase the capacitance and reduce the gap between GND and power plane. So high speed signals on top layer will have a solid GND reference plane which helps to reduce EMC emissions, as going up in number of layers and having a GND reference for each PCB signal layer will improve further the radiated EMC performance.

Figure 21. Four layer PCB stack-up example

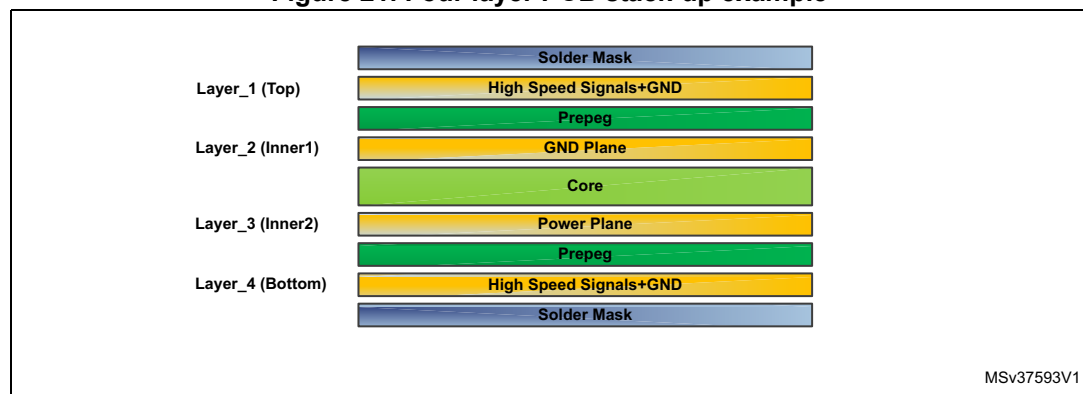
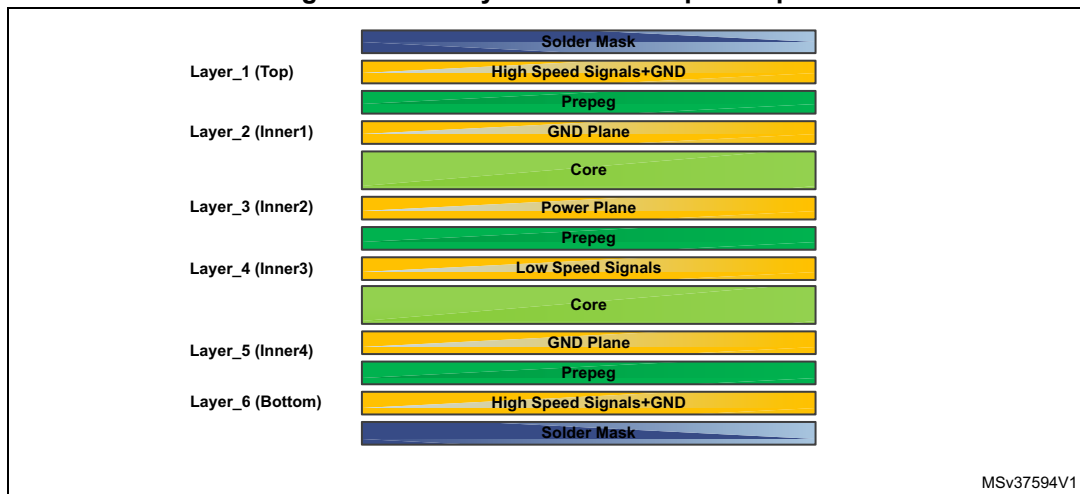


Figure 22. Six layer PCB stack-up example



8.2 Crystal oscillator

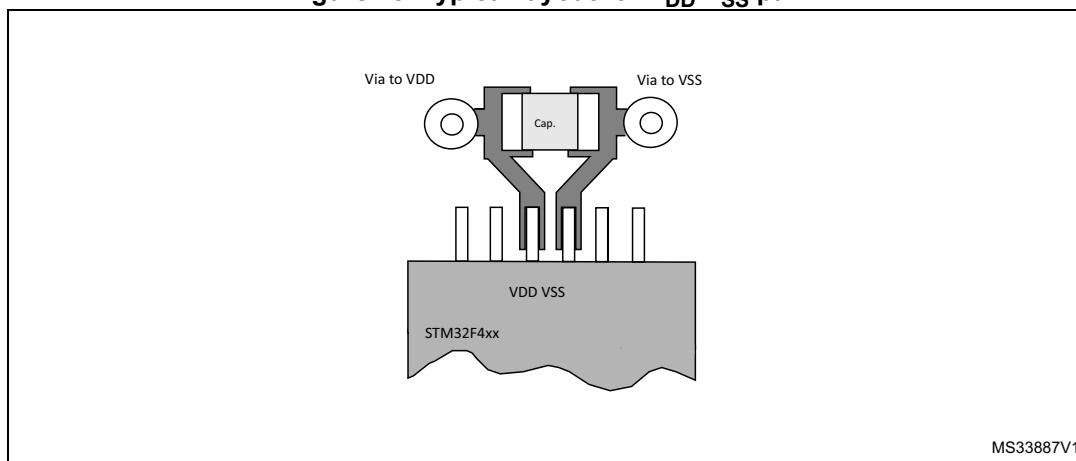
Use the application note: Oscillator design guide for STM8S, STM8A and STM32 microcontrollers (AN2867), for further guidance on how to layout and route crystal oscillator circuits.

8.3 Power supply decoupling

All power supply and ground pins must be properly connected to the power supplies. These connections, including pads, tracks and vias should have as low impedance as possible.

This is typically achieved with thick track widths and, preferably, the use of dedicated power supply planes in multilayer PCBs.

In addition, each power supply pair should be decoupled with filtering Ceramic capacitors (100 nF) and one single Tantalum or Ceramic capacitor (min. 4.7 μ F typ. 10 μ F) connected in parallel. These capacitors need to be placed as close as possible to, or below, the appropriate pins on the underside of the PCB. Typical values are 10 nF to 100 nF, but exact values depend on the application needs. Figure 22 shows the typical layout of such a VDD/VSS pair.

Figure 23. Typical layout for V_{DD}/V_{SS} pair

8.4 High speed signal layout

8.4.1 SDMMC bus interface

Interface connectivity

The SD/SDIO MMC card host interface (SDMMC) provides an interface between the APB2 peripheral bus and Multi Media Cards (MMCs), SD memory cards and SDIO cards. The SDMMC interface is a serial data bus interface, that consists of a clock (CK), command signal (CMD) and 8 data lines (D [0:7]).

Interface signal layout guidelines:

- Reference the plane using GND or PWR (if PWR, add 10nf switching cap between PWR and GND)
- Trace the impedance: $50\Omega \pm 10\%$
- The skew being introduced into the clock system by unequal trace lengths and loads, minimize the board skew, keep the trace lengths equal between the data and clock.
- The maximum skew between data and clock should be below 250 ps @ 10mm
- The maximum trace length should be below 120mm. If the signal trace exceeds this trace-length/speed criterion, then a termination should be used
- The trace capacitance should not exceed 20 pF at 3.3V and 15pF at 1.8V
- The maximum signal trace inductance should be less than 16nH
- Use the recommended pull-up resistance for CMD and data signals to prevent bus floating.
- The mismatch within data bus, data and CK or CK and CMD should be below 10mm.
- Keep the same number of vias between the data signals

Note: *The total capacitance of the SD memory card bus is the sum of the bus master capacitance.*

CHOST, the bus capacitance CBUS itself and the capacitance CCARD of each card connected to this line. The total bus capacitance is $CL = CHost + CBus + N * CCard$ where Host is STM32F4xxx, bus is all the signals and Card is SD card.

8.4.2 Flexible memory controller (FMC) interface

Interface connectivity

The FMC controller and in particular SDRAM memory controller which has many signals, most of them have a similar functionality and work together. The controller I/O signals could be split in four groups as follow:

- An address group which consists of row/column address and bank address.
- A command group which includes the row address strobe (NRAS), the column address strobe (NCAS), and the write enable (SDWE).
- A control group which includes a chip select bank1 and bank2 (SDNE0/1), a clock enable bank1 and bank2 (SDCKE0/1), and an output byte mask for the write access (DQM).
- A data group/lane which contains 8 signals (a): the eight D (D7–D0) and the data mask (DQM).

Note: It depends of the used memory: SDRAM with x8 bus widths have only one data group, while x16 and x32 bus-width SDRAM have two and four lanes, respectively.

Interface signal layout guidelines:

- Reference the plane using GND or PWR (if PWR, add 10nf stitching cap between PWR and GND)
- Trace the impedance: $50\Omega \pm 10\%$
- The maximum trace length should be below 120mm. If the signal trace exceeds this trace-length/speed criterion, then a termination should be used.
- Reduce the crosstalk, place data tracks on the different layers from the address and control lanes, if possible. However, when the data and address/control tracks coexist on the same layer they must be isolated from each other by at least 5 mm.
- Match the trace lengths for the data group within ± 10 mm of each other to diminish the skew. Serpentine traces (back and forth traces in an “S” pattern to increase trace length) can be used to match the lengths.
- Placing the clock (SDCLK) signal on an internal layer, minimizes the noise (EMI).
- Route the clock signal at least 3x of the trace away from others signals. Use as less vias as possible to avoid impedance change and reflection. Avoid using serpentine routing.
- Match the clock traces to the data /address group traces within ± 10 mm.
- Match the clock traces to each signal trace in the address and command groups to within ± 10 mm (with maximum of ≤ 20 mm).
- Trace the capacitances:
 - At 3.3 V keep the trace within 20 pF with overall capacitive loading (including Data, Address, SDCLK and Control) no more than 30pF.
 - At 1.8 V keep the trace within 15 pF with overall capacitive loading (including Data, Address, SDCLK and Control) no more than 20pF.

8.4.3 Quadrature serial parallel interface (Quad SPI)

Interface connectivity

The QUADSPI is a specialized communication interface targeting single, dual or Quad SPI

FLASH memories. The QUAD SPI interface is a serial data bus interface, that consists of a clock (SCLK), a chip select signal (nCS) and 4 data lines (IO[0:3]).

Interface signal layout guidelines

- Reference the plane using GND or PWR (if PWR, add 10nf stitching cap between PWR and GND)
- Trace the impedance: $50\Omega \pm 10\%$
- The maximum trace length should be below 120mm. If the signal trace exceeds this trace-length/speed criterion, then a termination should be used
- Avoid using multiple signal layers for the data signal routing.
- Route the clock signal at least 3x of the trace away from other signals. Use as less vias as possible to avoid the impedance change and reflection. Avoid using a serpentine routing.
- Match the trace lengths for the data group within ± 10 mm of each other to diminish skew. Serpentine traces (back and forth traces in an “S” pattern to increase trace length) can be used to match the lengths.

Avoid using a serpentine routing for the clock signal and as less via(s) as possible for the whole path. a via alter the impedance and add a reflection to the signal.

8.4.4 Embedded trace macrocell (ETM)

Interface connectivity

The ETM enables the reconstruction of the program execution. The data are traced using the data watchpoint and trace (DWT) component or the instruction trace macrocell (ITM) whereas instructions are traced using the embedded trace macrocell (ETM). The ETM interface is synchronous with the data bus of 4 lines D [0:3] and the clock signal CLK.

Interface signals layout guidelines

- Reference the plane using GND or PWR (if PWR, add 10nf stitching cap between PWR and GND)
- Trace the impedance: $50\Omega \pm 10\%$
- All the data trace should be as short as possible (≤ 25 mm),
- Trace the lines which should run on the same layer with a solid ground plane underneath it without a via.
- Trace the clock which should have only point-to-point connection. Any stubs should be avoided.
- It is strongly recommended also for other (data) lines to be point-to-point only. If any stubs are needed, they should be as short as possible. If longer are required, there should be a possibility to optionally disconnect them (e .g. by jumpers).

8.5 Package layout recommendation

8.5.1 BGA 216 0.8 mm pitch design example

Table 9. BGA 216 0.8 mm pitch package information

Package information (mm)	Design parameters (mm)
Ball pitch : 0.8	Via size : hole size $\varnothing = 0.2$, pad size: 0.45, plane clearance: 0.65
Ball size : 0.4	Trace width : 0.10/0.125
Number of rows/columns : 15x15	Trace/trace space : 0.10/0.125
Package solder Pad: SMD	BGA land size (Ball pad): $\varnothing = 0.4$, solder mask: 0.5

With 0.8 mm pitch BGA balls, fan-out vias are needed to route the balls to other layers on the PCB. Through-vias are used in this example, which cost less than blind, buried vias. For four adjacent BGA land pads, it is possible to have only one via as showing in [Figure 24](#) and [Figure 25](#). The traces are routed of two first row and two first colon without fan-out via. The current pitch size allows to route only one trace between two adjacent BGA land pads.

[Figure 26](#) shows an example of ideal SDRAM signals fan-out vias with power and gnd signals. These signals can be optimized to achieve the routing and length matching in an another layer before connecting to an SDRAM IC.

Figure 24. BGA 0.8mm pitch example of fan-out

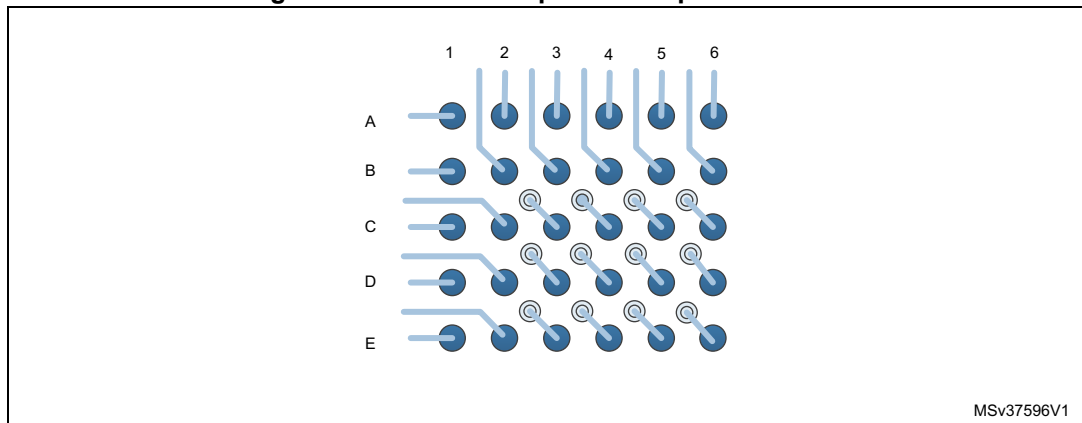


Figure 25. Via fan-out

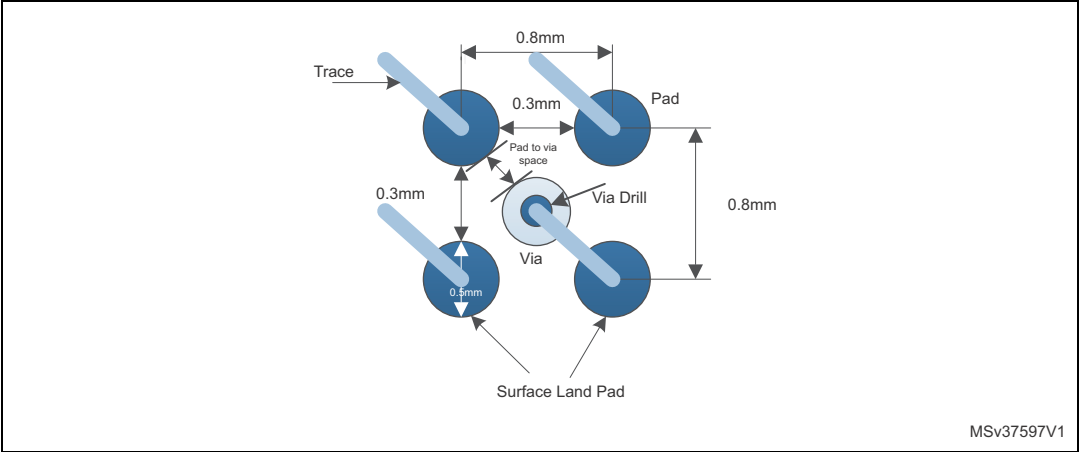
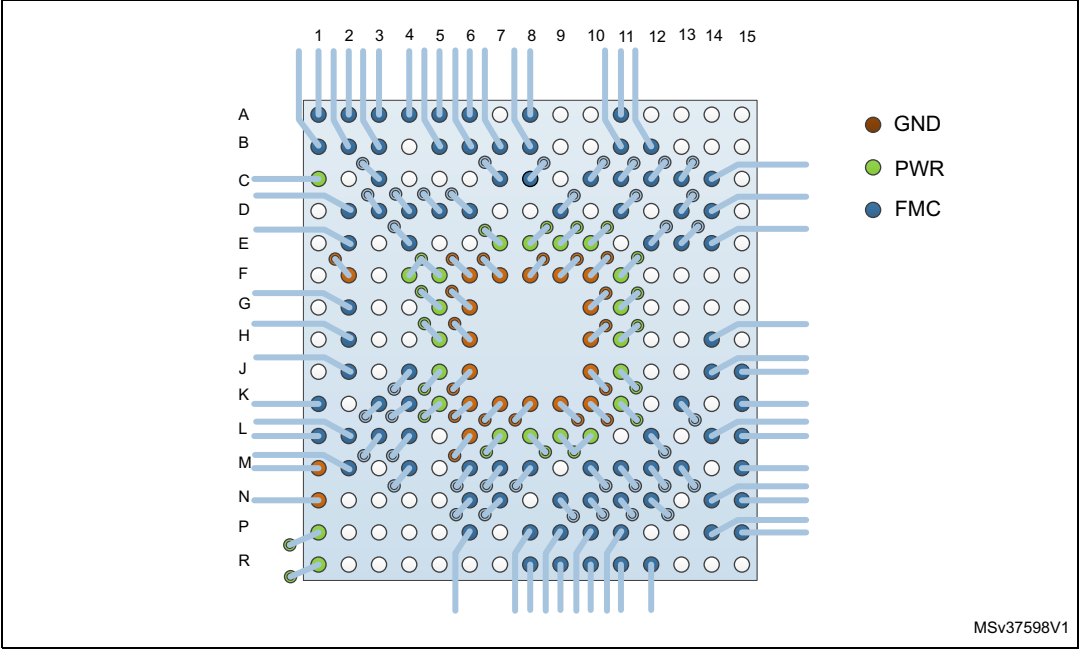


Figure 26. FMC signal fan-out routing example



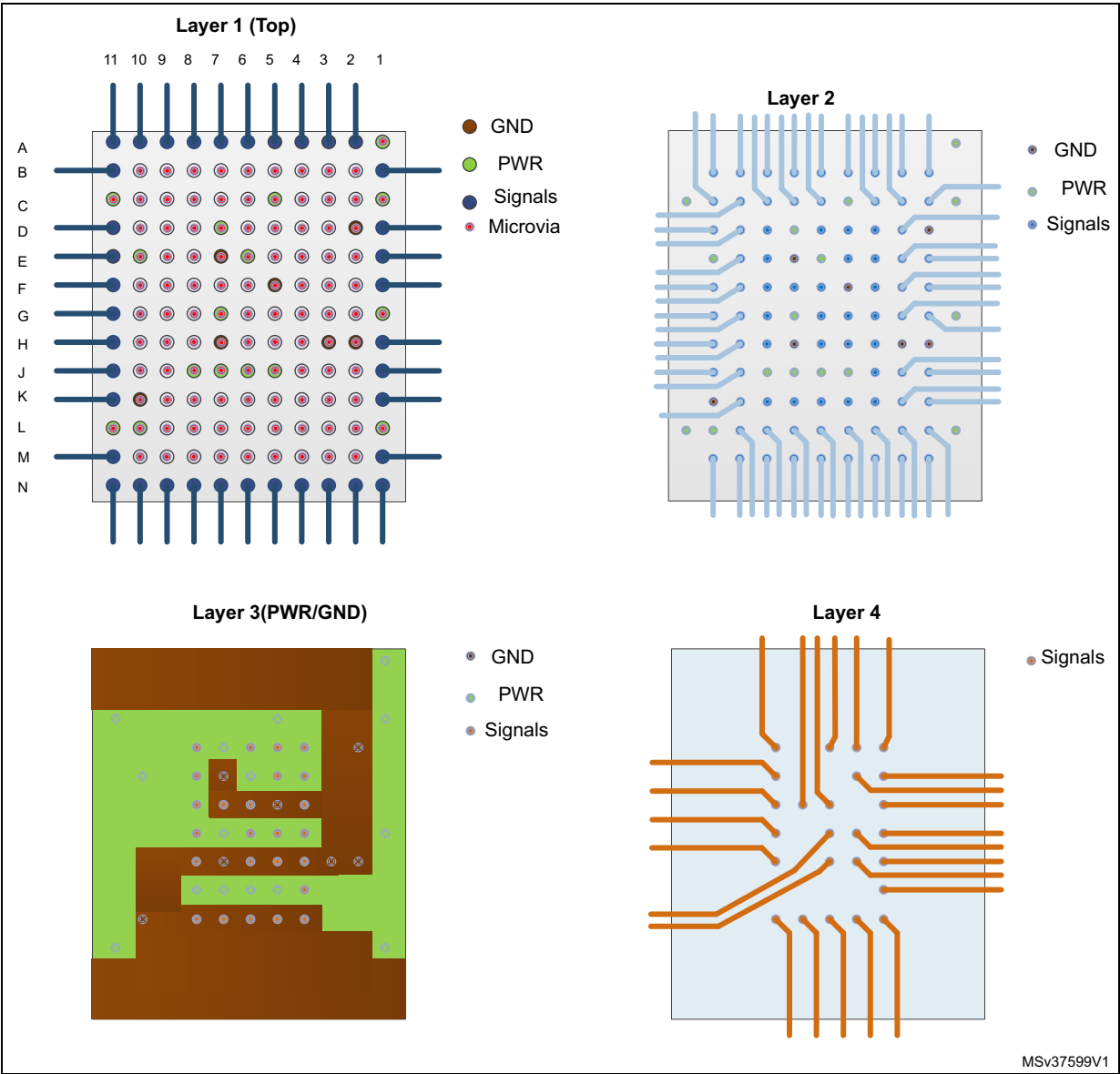
8.5.2 WLCSP143 0.4 mm pitch design example

Table 10. Wafer level chip scale package information

Package information (mm)	Design parameters (mm)
Bump pitch : 0.4	Microvia size : hole size $\varnothing = 0.1$, via land: 0.2
Bump size : 0.25	Trace width/space : 0.07/0.05 or 0.07/0.07
Number of rows/columns : 13x11	Bump pad size $\varnothing = 0.26$ max – 0.22 recommended
Non-solder mask defined via underbump allowed	Solder mask opening bump $\varnothing = 0.3$ min (for 0.26 diameter pad)

A better way to route this package and the fan-out signals is to use a through microvia technology. Microvia will route out internal bumps to a buried layers inside the PCB. To achieve this, the WLCSP package pads have to be connected to this internal layer through microvia. In case of four layers PCB, the first layer is WLCSP component, the second layer will be used as a signal layer, the third layer as the power and ground and the bottom layer for a signal layout. [Figure 27](#) shows an example of the layout for four layers PCB.

Figure 27. 143-bumps WLCSP, 0.40 mm pitch routing example



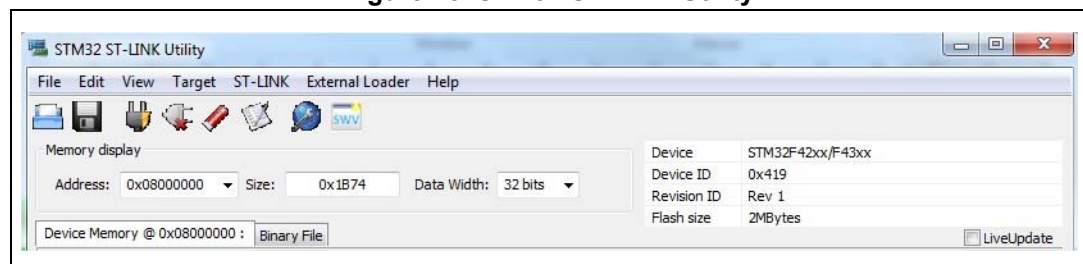
9 FAQ

9.1 Identify the STM32F4xxxx

In order to identify the STM32F4 refer to section “Part numbering” in your product datasheet.

To get the MCU's ID you can use ST-LINK Utility, once connected, the tool identify the target, and shows the ID, sub family, revision and flash size of the device as shown below.

Figure 28. STM32 ST-LINK Utility



9.2 Hardware tools available

ST provides three different development platforms:

9.2.1 Nucleao Boards

These boards have a wide extension capabilities with specialized shields like Arduino™ Uno.

Nucleao boards can easily be expanded through a variety of add-on boards.

9.2.2 Discovery kits

Discovery boards helps user to discover the high performance microcontrollers of the STM32 F4 series and to develop applications easily.

Different discovery boards are available. For example the STM32F429I-DISCO includes these features:

- SDRAM 64Mbits
- L3GD20, MEMS motion sensor, 3 axis digital output gyroscope
- USB OTG with micro AB connector...

9.2.3 Evaluation boards

Evaluation boards are a complete demonstration and development platform for the STM32 F4 series.

The full range of hardware features on the board is provided to help you evaluate all peripherals (USB OTG HS, USB OTG FS, Ethernet, motor control, CAN, MicroSD card, smartcard, USART, Audio DAC, RS-232, IrDA, SRAM, MEMS, EEPROM... etc.) and develop your own applications.

9.2.4 Where to find IBIS models?

IBIS models describe the electrical characteristics of the digital inputs and outputs of the STM32F4xxxx through I=f(V) and V=f(T) data.

For further information on IBIS models refer to AN4803 (High-speed SI simulations using IBIS and board-level simulations using HyperLynx SI on STM32 32-bit Arm® Cortex® MCUs)

IBIS models are available on ST's website.

9.3 MCU does not work properly

The table below summarizes some possible root causes that prevents the MCU from starting correctly. It presents also the most common suspects related to the possible root cause that you may need to verify.

Table 11. MCU does not work properly

Possible root cause	What you need to verify
VDD and GND feeding the device	- Power delivered to the device must be stable. - Monitor power supplies. - Check if GND is properly supplied to device. - Verify the GND coupling.
OSC_OUT	- Monitor OSC_OUT with oscillator to verify if it is working properly. - Refer to the AN2867: Oscillator design guide for ST microcontrollers.
RESET pin	- Check if RESET pin is correctly driven. - NRST connection includes a 100nF capacitor to ground.
BOOT PINs	- Monitor boot pins. - MCU may not work properly if BOOT0 and BOOT1 are floating.
VCAPs	VCAP1 and VCAP2 should be connected to 2x2.2 µF LowESR < 2 Ω Ceramic capacitor (or one 4.7 µF LowESR < 1 Ω Ceramic capacitor if only VCAP1 pin is provided on some packages).
PVD	- When programming the MCU, verify PLS [2:0] option bits. (you can use ST Link utility for that) - An interruption is generated depending of the programmed PVD threshold.
Temperature range	Respect worst conditions of temperature guaranteed by ST in the Datasheet.
System RESET	Care must be taken when PDR_ON pin =1, in such configuration the PDR block is active, that means if VDD drops down to 1.74 Volts (Refer to STM32F4xxxx datasheets for actual value) a system reset will occur.

10 Conclusion

This application note should be used as a starting reference for a new design with STM32F4xxxx device.

11 Revision history

Table 12. Document revision history

Date	Revision	Changes
20-Jun-2014	1	Initial release.
28-Oct-2014	2	Added STM32F411xC/xE in Table 1 Added footnote in Table 3 Updated Table 5 and Table 11 Updated Figure 2 , Figure 7 and Figure 8 Updated Section 3.2.1 Added Section 2.3.5 for STM32F411xC/xE Added Figure 5 and Figure 6
20-Mar-2015	3	Updated Table 1: Applicable products ; Updated Table 3: Regulator ON/OFF and internal power supply supervisor availability , Table 3: Package summary , Table 6: WLCSP Package summary , Table 5: Pinout summary and Table 11: Reference connection for all packages ; Updated Figure 9: STM32F4 family compatible board design for LQFP64 package , Figure 10: STM32F4 family compatible board design for LQFP100 package , Figure 12: Compatible board design STM32F10xx/STM32F4xx for LQFP64 package , Figure 13: Compatible board design STM32F10xx/STM32F2xx/STM32F4xx for LQFP100 package , Figure 14: Compatible board design STM32F10xx/STM32F2xx/STM32F4xx for LQFP144 package ; Added Figure 11: Compatible board design STM32F4xx / STM32F446xx for LQFP144 package .
21-Aug-2015	4	Updated – Figure 2: Power supply scheme (excluding STM32F469xx/F479xx) – Figure 3: Power supply scheme for STM32F469xx/F479xx and related notes . – Table 1: Applicable products ; – Table 3: Regulator ON/OFF and internal power supply supervisor availability – Table 3: Package summary , – Table 6: WLCSP Package summary – Table 17: Reference connection for all packages Added – Note 2 in Figure 12: Boot mode selection implementation example

Table 12. Document revision history (continued)

Date	Revision	Changes
07-Oct-2016	5	Added: – STM32F410x8, STM32F410xB, STM32F412xE, STM32F412xG, STM32F446xx, STM32F467xx and STM32F469xx. – Section 8: Recommended PCB routing guidelines for STM32F4xxxx devices – Section 9: FAQ – Figure 19: Reference schematic – Figure 20: Bill of Material – Figure 28: STM32 ST-LINK Utility Updated: – Section 7: Reference design
5-Dec-2016	6	Added: STM32F413/423 line Updated: – Table 1: Applicable products – Table 3: Package summary – Table 7: STM32F4xxxx bootloader communication peripherals
03-Oct-2018	7	Added: – Section 8.5: Package layout recommendation – Section 8.5.1: BGA 216 0.8 mm pitch design example – Section 8.5.2: WLCSP143 0.4 mm pitch design example Updated: – Section 1: Reference documents

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